

1 | A reusable experiment

A small calibrated offset is not enough. A comparator can decide the wrong way, miss its deadline, or change behavior after parasitic extraction.

This notebook makes the complete decision map, costs and failure boundaries reproducible instead of showing only one favorable nominal waveform.

Circuit and design choices

Established StrongARM topology; 27 transistor instances.
LVT main/auxiliary input path; standard-VT latch and switches.
Physical source-gated trim: sign plus four magnitude bits.
Nine declared candidates, followed by an explicit lower-energy control. No universal best-design claim.

Model applicability

The pinned Magic/open_pdks RC export retains mutual coupling while ground capacitance increases. Physical error size and direction are unknown. C-only is not independent ground truth; RC-versus-C differences are not pure resistance effects.

Open the current submission in Colab

Comparator_Atlas.ipynb

Run all: archived evidence and waveform measurements. DRC/LVS establish structural checks, not model fidelity. Schematic results are unaffected by this extraction concern. No corrected counts, energy or silicon PVT guarantee is inferred.

Archived RC-deck outcomes; model physical fidelity not yet qualified. Core energy excludes external drivers and calibration infrastructure.

Established circuitry: Razavi, 2015, DOI 10.1109/MSSC.2015.2418155; Li, Xu & Iizuka, 2022, DOI 10.1007/s10470-022-01992-6.

Original code: MIT. SKY130 keeps Apache-2.0. GitHub Copilot assistance disclosed. Source, models and reproduction instructions are linked in the notebook.

2 | Schematic coverage is not free

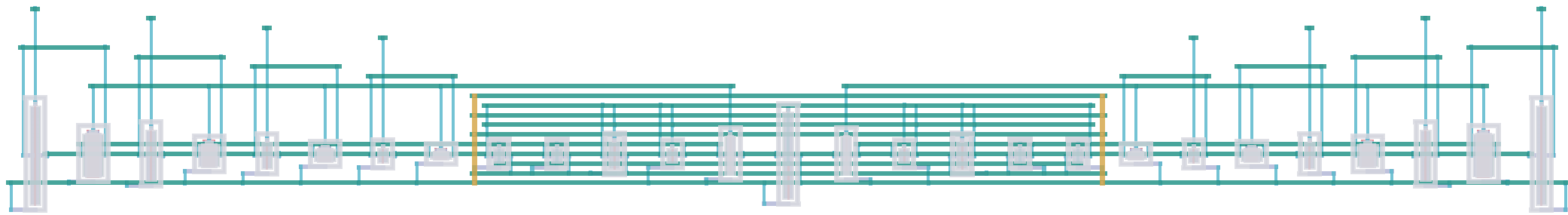
Same local calibration; 1 ns; sampled |input| >= 1 mV; +4% branch-width stress



All 49 conditions: 79.1% -> 94.9% correct | 143.0 -> 251.5 fJ/cycle

Above: 45 PVT combinations; aggregate also includes four nominal stress controls. Percentages are grid coverage, not yield.

3 | Structural checks and archived-deck outcomes



Actual 27-device GDS | DRC 0 + LVS / negative controls | 2207.088 um2

Archived 45-PVT RC: 156/180 at 1 ns | 180/180 at declared 2 ns

Worst sampled RC decision: 1.835 ns at FS / 1.62 V / -40 C / -3 mV. Four signed inputs per condition; 24 unresolved at 1 ns. Separate ngspice-42 pilot: 12/20 at failed original 1 ns; 20/20 at post-hoc 2 ns. No continuous-input or statistical-yield guarantee.

Matched 180-point mean core energy, same ngspice-47 conditions:

Schematic: 245.8 fJ | Extracted RC: 425.5 fJ